

Understanding the Network to Be Monitored

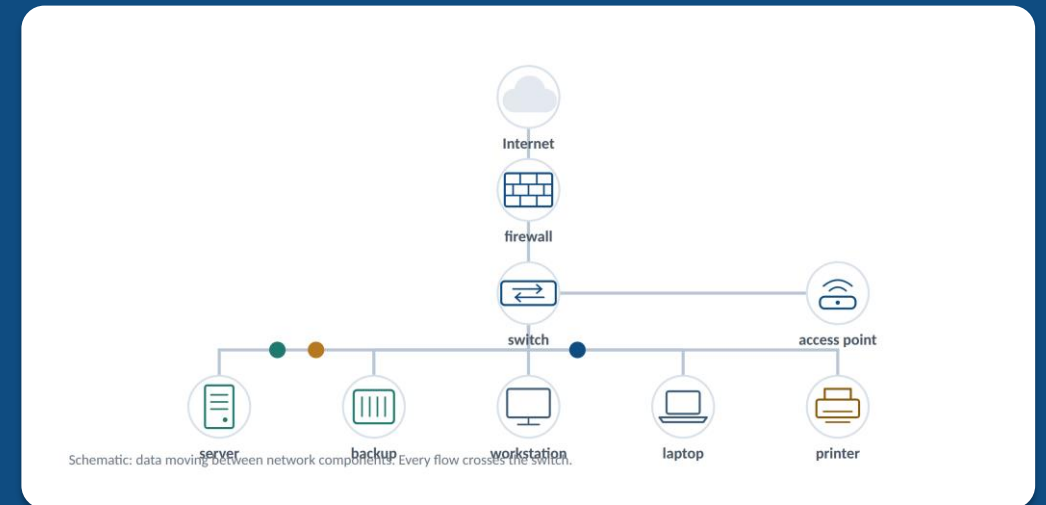
Lesson 2.1 · Network components

Lesson 2.2 · Network documentation and topology diagrams

Monday, September 21, 2026 · online · 60 minutes

Asen Grozdanov · Cyber.SoHo Educational Hub

LEA.D8 (Cybersecurity: Prevention and Intervention) program · Dawson College



Schematic: data moving between the parts of a small network. Every flow crosses the switch.

Tonight: from principles to a real network



Where we are

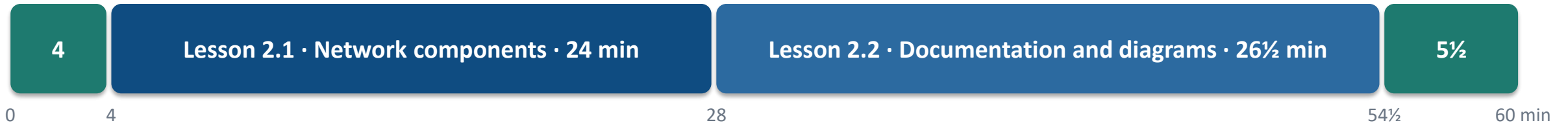
- Topic 1 gave you the principles: the **CIA** (Confidentiality, Integrity, Availability) triad, the main types of attack, prevention, detection and response, and security versus performance issues.
- Topic 2 starts the practical work: before monitoring a network, you must know what is on it and how it is connected.
- Competency **EZ01** (Monitor a computer network from a cybersecurity perspective), element 2: analyze the topology of the network.



By the end of this hour you will be able to

1. Classify any device as a server, a workstation, network equipment or another endpoint, by its **role**.
2. Explain what each family does and which data we collect from it.
3. Read a physical and a logical network diagram.
4. Follow a data flow and find the monitoring points and the single points of failure.
5. Recognize outdated documentation and know what to do about it.

Tonight's plan: 60 minutes



Opening

- Objectives
- Plan
- Rules for the material



Lesson 2.1

- What a network is made of
- Servers, workstations, other endpoints
- Network equipment
- Example, code, check yourself



Lesson 2.2

- Documentation, physical and logical diagrams
- Addresses and topologies
- A reading method
- Example, code, check yourself



Wrap-up

- Lab 01 preview
- Key points
- Next class and links



Everything on these slides is also in the lecture page (the two step-by-step code players are there) and in the download package.



Rules for tonight's material



Everything tonight is fictional

- The **Fictiva Dental Clinic** does not exist; its hostnames are invented.
- Internal addresses come from the private ranges of **RFC** (Request for Comments) 1918.
- The public address comes from a documentation range of **RFC** (Request for Comments) 5737.
- The domain `fictiva-dental.example` uses a name reserved by **RFC** (Request for Comments) 2606.
- We only read documents and run two small scripts on text files: nothing touches a real network.



Legal and ethical frame

- Real network documentation is confidential. Never use an employer's, a client's or a relative's diagrams without written permission (Legal Notice, sections 6.7, 7.5 and 7.9).
- Run the scripts only in your Authorized Lab Environment (section 6.4).
- Full text of the rules:
 - [Cyber.SoHo Legal Notice, version 2.0](#)



Before opening any link, check it with [VirusTotal](#) (Legal Notice, Annex C).

Lesson

2.1

Network components

Servers, workstations and network equipment: what each family does, how to recognize it, and which data we collect from it.

The family is decided by the role, not by the appearance.



Servers

Provide services to other devices; nobody works on them directly.

At the clinic: File server, backup storage



Workstations

Used directly by one person, with a full operating system.

At the clinic: Reception desktop, office laptop



Network equipment

Forward traffic that belongs to other devices.

At the clinic: Switch, firewall, access point, modem



Other endpoints

Fixed-function devices that run firmware only.

At the clinic: Printer, entrance camera



What a network is made of



A **computer network** is a set of devices, called **nodes**, connected by **links** (cables or radio) so that they can exchange data.



A node that has its own network address and runs programs that send or receive data is a **host**.



From a monitoring point of view, every node is an **asset**: it has value for the organization and must be protected.



The **asset inventory** lists every asset with its **hostname**, address, role, location, owner and dependent devices.

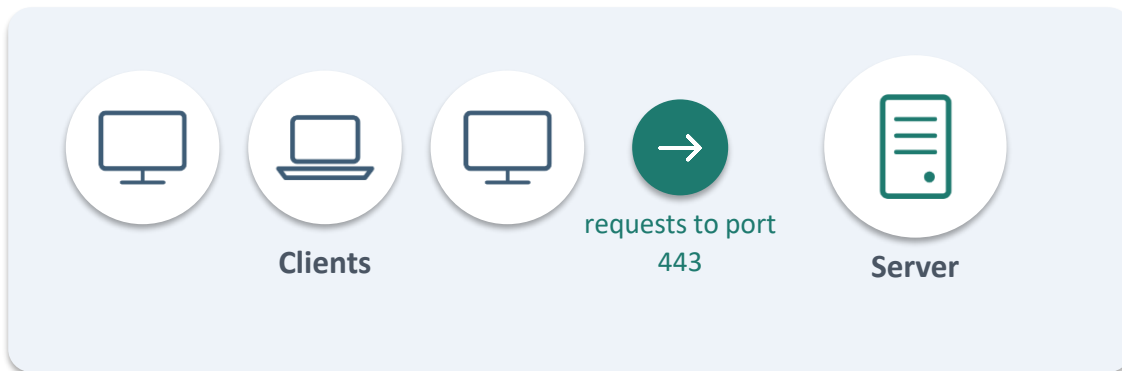


Why the inventory comes first

- A monitoring tool watches only the devices it has been told about. A missing device is not monitored: no alert if it fails or is attacked.
- Control 1 of the **CIS** (Center for Internet Security) Controls:
 - [Inventory and Control of Enterprise Assets](#)
- **NIST** (National Institute of Standards and Technology) Cybersecurity Framework: asset management, category **ID.AM** (Identify function, Asset Management category), opens the Identify function.
 - [The framework's official page](#)
- Canadian Centre for Cyber Security:
 - [Baseline controls for small organizations](#)

Servers (1/2)

- A **server** is a computer, physical or virtual, whose main function is to provide a **service** to other computers, its **clients**.
- This arrangement is the **client-server model**.
- A service is a program that runs continuously and waits for requests on a numbered **port**. Web services usually wait on port 443 of the **TCP** (Transmission Control Protocol).
- One server can run several services, and a service can move to another server.



Common server roles in a small organization



File server
Shared folders



Database server
Structured records, such as patient files



DNS (Domain Name System) server
Translates names into addresses



DHCP (Dynamic Host Configuration Protocol) server
Gives each new device an address



Backup server or NAS (Network-Attached Storage)
Keeps copies of the data



Web and email servers
Publish pages and deliver messages

Servers (2/2)

Built to run 24 hours a day



ECC (Error-Correcting Code) memory detects and corrects single-bit memory errors.



Several disks combined with **RAID** (Redundant Array of Independent Disks): one failed disk does not stop the server.



Two power supplies, and a separate remote management interface.



Many servers are **virtual machines**: one physical host runs several servers and becomes a **single point of failure** for all of them.



What we monitor on a server

- Does each service answer requests?
- Processor, memory and disk use
- Successful and failed logons, account changes (event logs)
- The result of every backup

A server concentrates data (confidentiality) and services that many people need (availability).

A client-server exchange at the clinic

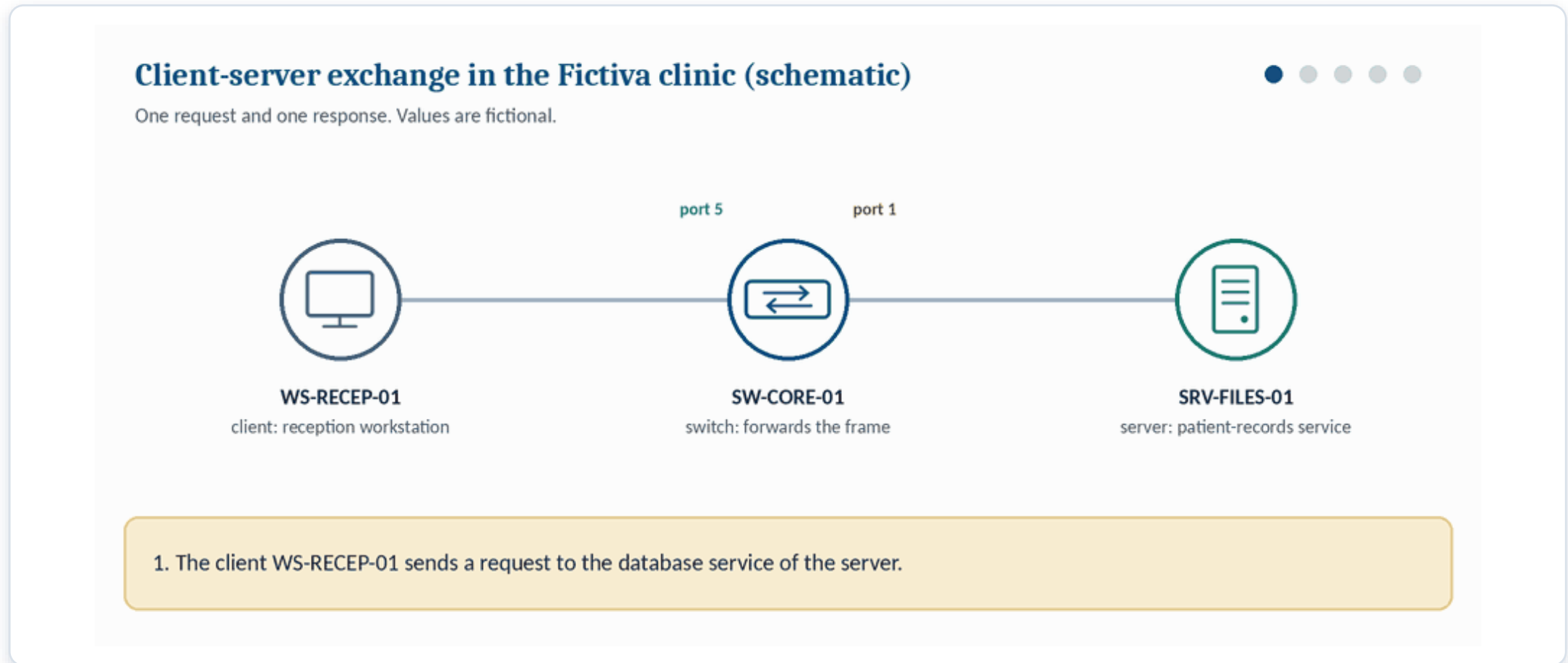


Figure 2.1-A. The reception workstation sends a request, the switch forwards it to the server's port only, the database service answers. The last frame lists what a monitoring tool watches on the server.

Server and workstation compared

Characteristic	Server	Workstation
Main function	Provides services to many clients	Used directly by one person
Who works on it	Nobody in daily use; administrators connect remotely	One user at a time, during working hours
Operating system	Server edition: Windows Server or a Linux server distribution	Desktop edition: Windows 11, macOS or a Linux desktop
Hardware built for	Continuous operation: error-correcting memory, disk redundancy, two power supplies	Comfort and cost: screen, keyboard, one disk, one power supply
Expected availability	24 hours a day, 7 days a week	Working hours; can be switched off at night
Typical risks	Service outage, data theft, ransomware on shared data	Phishing, malicious downloads, lost or stolen laptop
What we monitor	Service availability, resource use, logons, backups, configuration changes	Security events, failed logons, protection alerts, update level



The role decides the family: the same hardware can be a server in one office and a workstation in another.



Workstations and other endpoints



Workstations

- An **endpoint** used directly by one person: a desktop or a laptop with a general-purpose **OS** (Operating System) (Windows, macOS, Linux).
- Where people act (email, web, **USB** (Universal Serial Bus) keys): the usual starting point of phishing and malware.
- Laptops leave the building and join other networks.
- **We monitor:** security events, failed logons, protection alerts, update level.



Other endpoints

- Printers, scanners, **IP** (Internet Protocol) cameras, door controllers and other **IoT** (Internet of Things) devices.
- They run **firmware**: software stored in the device and designed for one task.
- Easy to forget, rarely updated, often left with their factory settings.
- **We monitor:** reachability, firmware version, unexpected connections.



Camera images and patient files are personal information, protected in Québec by [Law 25](#), supervised by the **CAI** (Commission d'accès à l'information du Québec). Example 2.1 gives them extra priority points.

Network equipment (1/2)

Layer 3 · Network

Packets between networks · **IP** (Internet Protocol) addresses

Layer 2 · Data link

Frames inside one local network · **MAC** (Media Access Control) addresses

Layer 1 · Physical

Signals on the cable or the radio channel

Three of the seven layers of the **OSI** (Open Systems Interconnection) model.

The switch

- **Network equipment** forwards traffic that belongs to other devices.
- Every node has a **NIC** (Network Interface Card) with its own **MAC** (Media Access Control) address.
- A **switch** connects the devices of one **LAN** (Local Area Network). It learns which address is behind which port and keeps the pairs in its **MAC (Media Access Control) address table**.
- It forwards each frame only to the destination's port; a frame for an unknown address goes to all ports.
- A **managed switch** can copy the traffic of chosen ports to another port: **port mirroring**, or **SPAN** (Switched Port Analyzer). Lab 02 uses it.
- The older **hub** repeated every signal to all ports; it is no longer installed.



Network equipment (2/2)



Router

Connects different networks, such as the clinic's **LAN** (Local Area Network) and the provider's **WAN** (Wide Area Network). It reads the destination **IP** (Internet Protocol) address and uses its **routing table**. For local devices it is the **default gateway**.



Firewall

Allows or blocks traffic with rules on addresses, ports and applications, and records its decisions in logs. It sits on the boundary between trusted and untrusted networks, and often also routes and performs **NAT** (Network Address Translation).



Wireless access point

Connects Wi-Fi devices, which follow the **IEEE** (Institute of Electrical and Electronics Engineers) 802.11 standards, to the wired Ethernet network (802.3). It can broadcast several network names, each an **SSID** (Service Set Identifier): staff and guests.



Modem

Converts the signal of the provider's line (fibre, cable or telephone) into Ethernet. In **bridge mode** it passes traffic without routing it. It usually belongs to the provider.



Small offices often use one **SOHO** (Small Office/Home Office) box as router, firewall, switch, access point and **DHCP** (Dynamic Host Configuration Protocol) server: document each function separately.

How a switch and a router decide

How network equipment decides where to send data

Left: a switch reads MAC (Media Access Control) addresses. Right: a router reads IP (Internet Protocol) addresses. OSI = Open Systems Interconnection model.

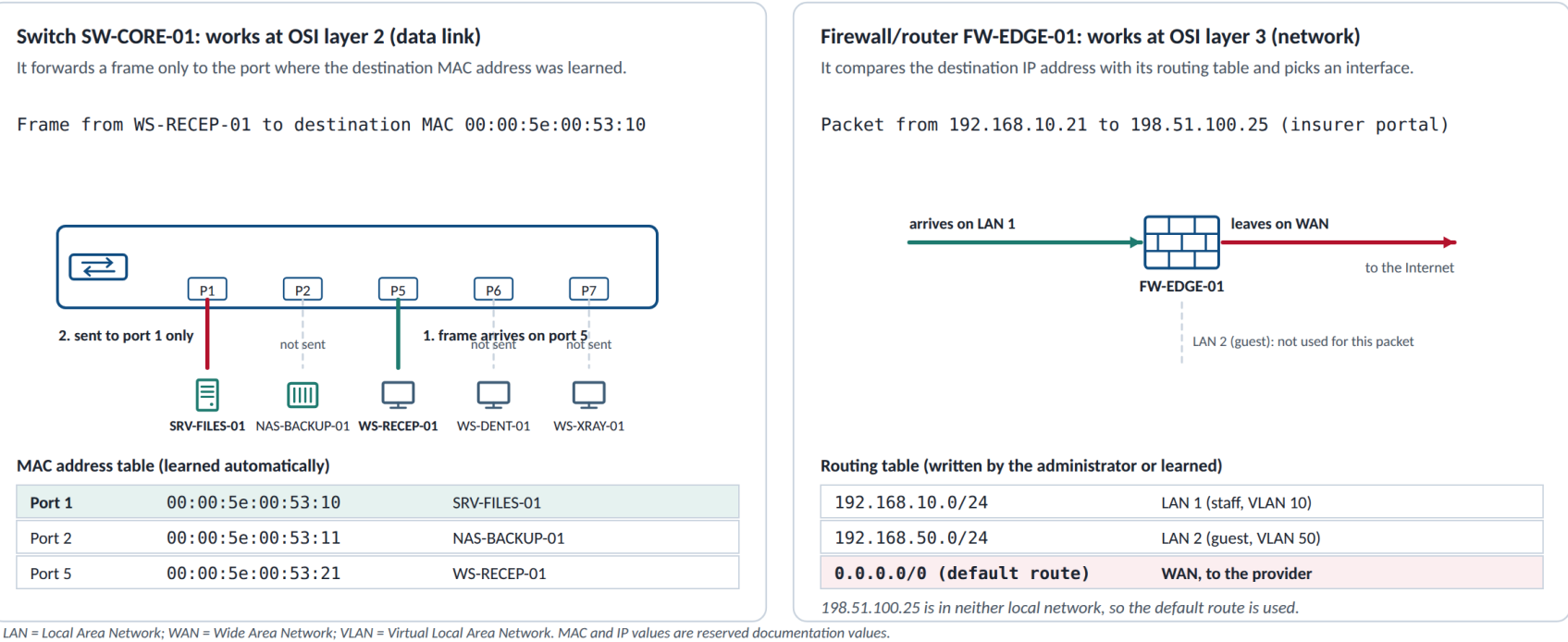


Figure 2.1-B. The switch uses its **MAC** (Media Access Control) address table; the firewall/router uses its routing table and the default route. Documentation addresses only (**RFC** (Request for Comments) 5737 and **RFC** (Request for Comments) 7042).



Network equipment reference

Device	What it connects	Decides with	Layer	What we monitor
Network interface card	One node to the network medium	Its own MAC (Media Access Control) address	1 and 2	Link status, errors (through the host)
Hub (no longer installed)	Devices of one LAN (Local Area Network)	Nothing: repeats every signal to every port	1	Replace it with a switch
Switch	Devices of one LAN (Local Area Network)	Destination MAC (Media Access Control) address and its address table	2	Port status, errors, traffic per port, logs
Wireless access point	Wi-Fi devices to the wired network	MAC (Media Access Control) addresses (802.11 to 802.3)	2	Connected and unknown devices, signal, logs
Router	Different networks, such as LAN (Local Area Network) to WAN (Wide Area Network)	Destination IP (Internet Protocol) address and its routing table	3	Interface status, routes, traffic volume, logs
Firewall	Trusted and untrusted zones	Rules on addresses, ports and applications	3 and above	Allowed and blocked connections, rule changes
Modem	Provider's line to Ethernet	Nothing: converts signals	1 and 2	Line up or down; the provider owns it



Every packet crosses network equipment: if it fails, everything behind it is cut off. We monitor it with **SNMP** (Simple Network Management Protocol) (status and counters), logs sent to a log server (Syslog) and copies of its traffic.

Videos for Lesson 2.1 (after class)



What is a Server? Servers vs Desktops Explained

PowerCert Animated Videos · 7 min 14 s

Watch for: the definition of a server by its role; the hardware differences with a desktop (processor, error-correcting memory, disk redundancy); the main server types.

Watch on YouTube



Hub, switch or router? Network devices explained

PieterExplainsTech · about 8 min

Watch for: how a switch fills its table from the source addresses of frames; why a router is needed between networks; the layer of each device.

[Criminal Code, section 184](#): capturing other people's traffic without authorization may be an offence.

Watch on YouTube



Links checked on September 20, 2026. Third-party content, no affiliation with Cyber.SoHo. Check each link with [VirusTotal](#) first.



Example 2.1: the clinic's first inventory



Task

Build the asset inventory of a small clinic: classify each device into a family and give it a monitoring priority.



Link to the lecture

Applies the four family definitions of Lesson 2.1 and the **CIA** (Confidentiality, Integrity, Availability) triad of Topic 1: availability (how many devices depend on it) and confidentiality (does it hold personal information).



Link to the labs

Lab 01 (Friday, September 25) starts with this deliverable for your network. **Lab 02** (Saturday, October 3) uses the priority order to decide what the tool watches first. **Lab 03** (Friday, October 16) uses the inventory to spot a device that should not be there.



The situation

The fictional clinic has two dentists, a hygienist, a receptionist and an office manager. The installer left a short documentation file.

The office manager asks you, the new monitoring technician:

“What do we have, and what should we watch first?”

The clinic's device list

Hostname	Address	What it is	Forwards traffic	Full operating system	Services offered	Person uses it	Personal information	Internet-facing	Dependents
MODEM-ISP-01	bridge mode	Provider's fibre modem	yes	no	0	no	no	yes	11
FW-EDGE-01	192.168.10.1	Firewall and router, gateway	yes	no	1	no	no	yes	10
SW-CORE-01	192.168.10.2	24-port managed switch	yes	no	0	no	no	no	9
AP-FRONT-01	192.168.10.3	Wi-Fi access point	yes	no	0	no	no	no	1
SRV-FILES-01	192.168.10.10	Patient records, shared folders	no	yes	2	no	yes	no	4
NAS-BACKUP-01	192.168.10.11	Backups, camera recordings	no	yes	1	no	yes	no	2
WS-RECEP-01	192.168.10.21	Reception desktop	no	yes	0	yes	yes	no	0
WS-DENT-01	192.168.10.22	Treatment room 1 desktop	no	yes	0	yes	no	no	0
WS-XRAY-01	192.168.10.24	X-ray imaging desktop	no	yes	0	yes	yes	no	0
LT-ADMIN-01	192.168.10.31	Office manager's laptop	no	yes	0	yes	no	no	0
PRN-FRONT-01	192.168.10.40	Printer and scanner	no	no	2	no	no	no	0
CAM-DOOR-01	192.168.10.41	Entrance camera	no	no	1	no	yes	no	0



Decide before the script runs: which family for NAS-BACKUP-01? For CAM-DOOR-01? Which device would you monitor first, and why?

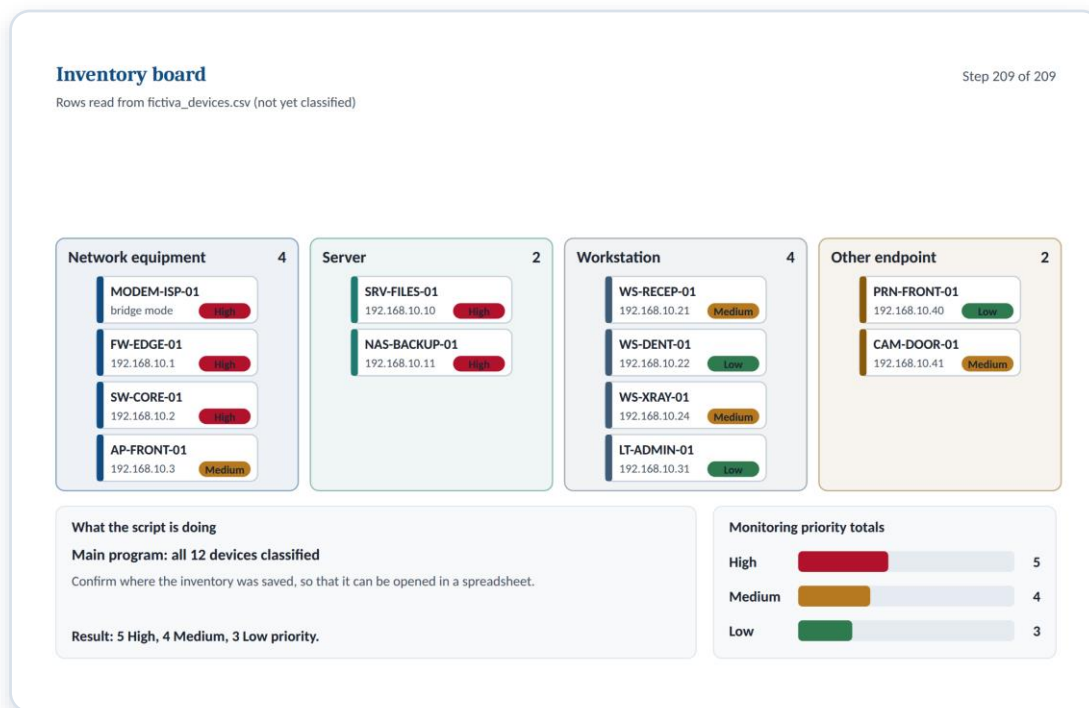
Code 2.1: asset_inventory.py

What the script does

1. Reads `fictiva_devices.csv`, one row per device.
2. `classify()` tests four rules in order; the first match gives the family.
3. `priority_score()` adds points: 4 or 2 for dependent devices, 2 for personal information, 2 for Internet exposure, 1 for a person using the device.
4. `priority_label()`: 4 points or more is High, 2 to 3 is Medium, 0 to 1 is Low.
5. Prints the inventory and saves `asset_inventory.csv`.

```
$ python3 asset_inventory.py
# Windows: py asset_inventory.py
```

Run it only in your lab environment, next to `fictiva_devices.csv`.



Live visual in the lecture page, last step: 4 network equipment, 2 servers, 4 workstations, 2 other endpoints. Start, Pause, Resume and Restart follow every line.

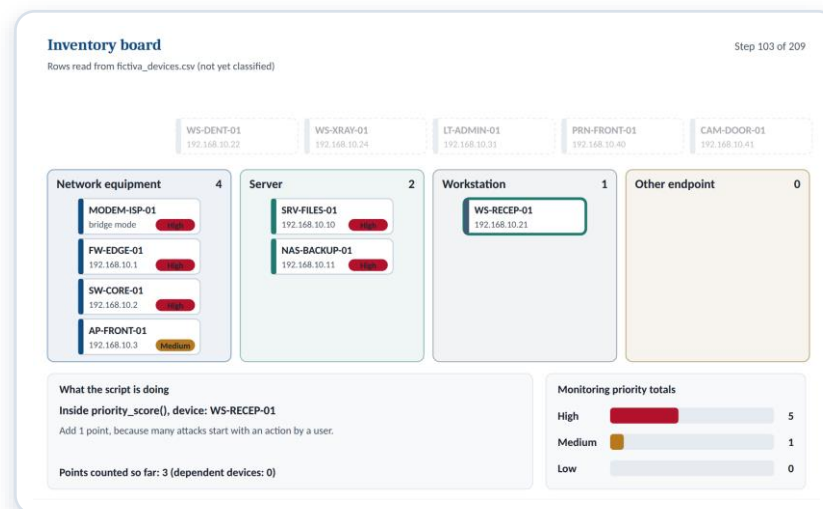
The classification rules in code

asset_inventory.py

```
def classify(device):
    # Rule 1: forwards other devices' traffic -> network equipment
    if device["forwards_traffic"] == "yes":
        return "Network equipment"
    # Rule 2: no full operating system -> other endpoint
    if device["full_os"] == "no":
        return "Other endpoint"
    # Rule 3: offers a service and nobody works on it -> server
    if (int(device["services_offered"]) > 0
        and device["interactive_user"] == "no"):
        return "Server"
    # Rule 4: a person uses it directly -> workstation
    if device["interactive_user"] == "yes":
        return "Workstation"
    # No rule matched: a person must check the documentation
    return "Unclassified"
```

Condensed excerpt. The full script, with a comment before every line, is in the lecture page and the download package.

- Rules are tested **in order**; the first match wins and return ends the function.
- MODEM-ISP-01: rule 1 matches at once.
- WS-RECEP-01: rules 1 to 3 are false; rule 4 matches.
- No match: “Unclassified”, a person checks.



Decision flow and priority points

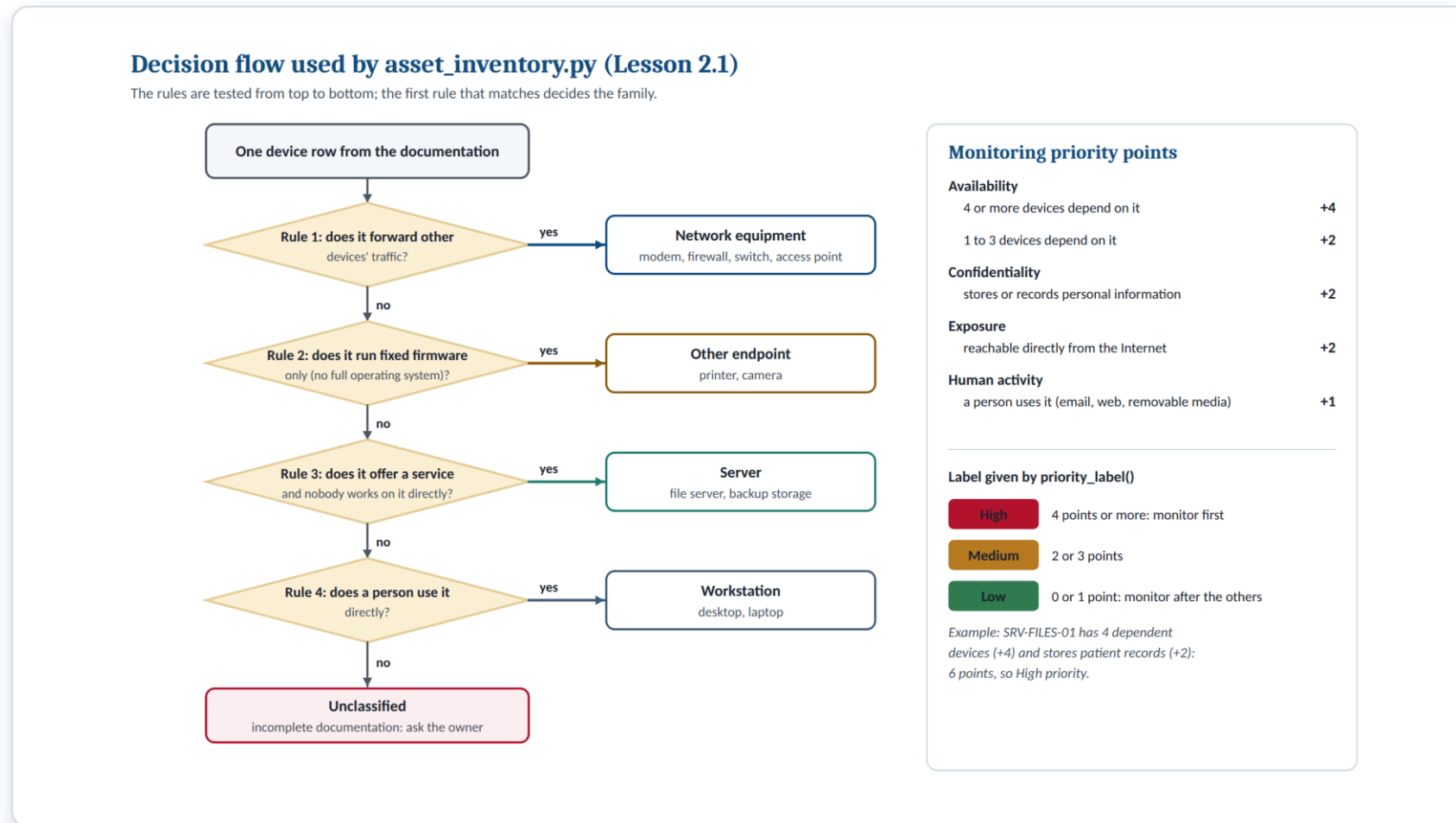
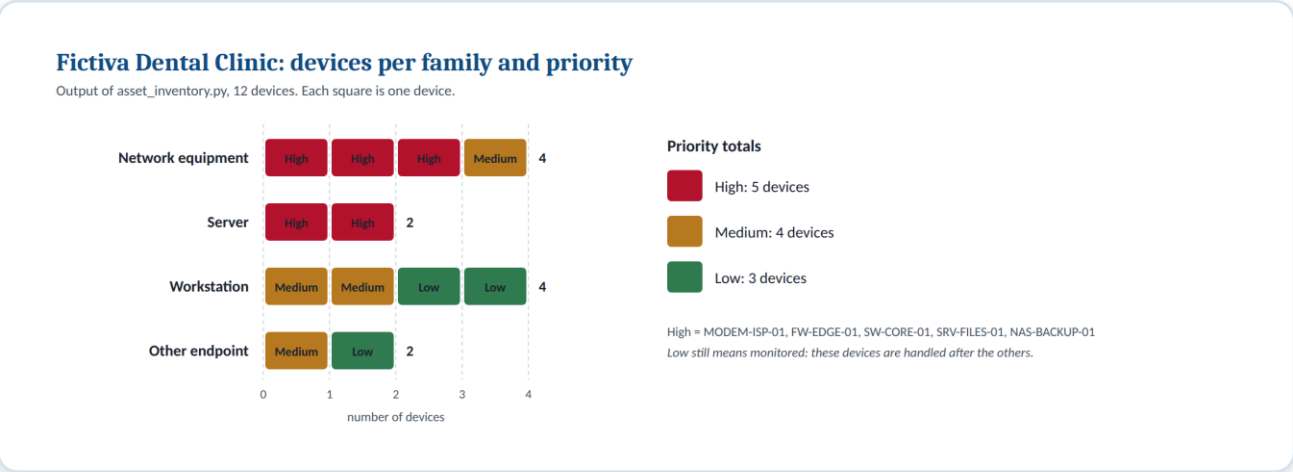


Figure 2.1-C. The rules of classify() and the points of priority_score(), drawn as a diagram you can reuse in Lab 01.

Result: the clinic's inventory



5
High priority

4
Medium priority

3
Low priority

Hostname	Family	Points	Priority
MODEM-ISP-01	Network equipment	6	High
FW-EDGE-01	Network equipment	6	High
SW-CORE-01	Network equipment	4	High
AP-FRONT-01	Network equipment	2	Medium
SRV-FILES-01	Server	6	High
NAS-BACKUP-01	Server	4	High
WS-RECEP-01	Workstation	3	Medium
WS-DENT-01	Workstation	1	Low
WS-XRAY-01	Workstation	3	Medium
LT-ADMIN-01	Workstation	1	Low
PRN-FRONT-01	Other endpoint	0	Low
CAM-DOOR-01	Other endpoint	2	Medium

 Fictiva_Asset_Inventory.xlsx recalculates by itself: change a “yes” to “no” and the family and priority update.

Check yourself: flash cards 2.1

1

What is a server?

A computer, physical or virtual, whose main function is to provide a service (files, database, names, addresses, web pages) to other computers, its clients. Its role defines it, not its hardware.

2

How do a workstation and a server differ?

A workstation is an endpoint used directly by one person. A server answers requests from many clients, usually without anyone working on it directly.

3

What does a switch use to forward frames, and at which layer does it work?

Destination **MAC** (Media Access Control) addresses, compared with the address table it learns from source addresses. Layer 2, data link.

4

What does a router do?

It connects different networks and forwards packets by comparing the destination **IP** (Internet Protocol) address with its routing table. Layer 3, network. It is the default gateway of its networks.

5

Why is the asset inventory the first step of preventative monitoring?

A monitoring tool watches only the devices it knows. The inventory says what exists, what each device does, who depends on it and in which order to monitor it.



Practice questions 2.1 (1/3)

Question 1 · one answer

At Fictiva, which device forwards packets between the staff network 192.168.10.0/24 and the Internet?

- a) SW-CORE-01, the switch
- b) FW-EDGE-01, the firewall and router
- c) AP-FRONT-01, the access point
- d) SRV-FILES-01, the server

Answer: b

Forwarding between networks is the router's job; at Fictiva the router function is inside FW-EDGE-01, the default gateway 192.168.10.1.

Question 2 · one answer

A computer runs the patient-records database and the shared folders. Nobody logs on to it and eight computers use it. Which family?

- a) Workstation
- b) Network equipment
- c) Server
- d) Other endpoint

Answer: c

It provides services (database, file sharing) to clients over the network. The role decides the family, whatever the hardware.



Practice questions 2.1 (2/3)

Question 3 · one answer

Which set of features is typical of server hardware and not of an ordinary office desktop?

- a) A large screen, a webcam and speakers
- b) Error-correcting memory, disk redundancy and two power supplies
- c) A wireless card and a battery
- d) A web browser and an email program

Answer: b

These features keep the server running when a memory bit, a disk or a power supply fails, because many people depend on it.

Question 4 · one answer

A switch receives a frame whose destination MAC (Media Access Control) address is already in its address table, on port 1. What does it do?

- a) It sends the frame to every port
- b) It sends the frame only to port 1
- c) It sends the frame to the default gateway
- d) It drops the frame

Answer: b

The table tells the switch where the destination was learned, so the frame leaves by that port only. Sending to every port is what a hub did.



Practice questions 2.1 (3/3)

Question 5 · 4 answers

Which of these devices are network equipment? Select all that apply.

- a) SW-CORE-01, the switch
- b) FW-EDGE-01, the firewall and router
- c) AP-FRONT-01, the access point
- d) NAS-BACKUP-01, the storage
- e) MODEM-ISP-01, the modem
- f) PRN-FRONT-01, the printer

Answer: a, b, c, e

Ask one question: does it forward other devices' traffic? The storage offers a service (server); the printer runs firmware for one task (other endpoint).

Question 6 · 4 answers

Which are monitoring data normally collected from a server? Select all that apply.

- a) Whether each service answers
- b) Processor, memory and disk use
- c) Logons in the event logs
- d) The Wi-Fi names in the building
- e) The result of the nightly backup
- f) Free ports on the switch

Answer: a, b, c, e

Keep what describes the server's availability, performance or security. Wi-Fi names concern access points; free ports concern the switch.



Lesson

2.2

Documentation and diagrams

The documents that describe a network, the two kinds of diagram, addresses, topologies, and a method to read any diagram.

A diagram without a date cannot be trusted.



2.2.1

What network documentation is



2.2.2

Physical and logical diagrams



2.2.3

Reading the addresses



2.2.4

Topologies



2.2.5

A method to read any diagram



2.2.6

When documents and reality disagree



What network documentation is

- **Network documentation** is the set of documents that describe a network as it is built and operated.
- Competency **EZ01** (Monitor a computer network from a cybersecurity perspective): monitoring is carried out “based on descriptive infrastructure documents”.
- When you join a team, it is how you learn the network. When an alert arrives at three in the morning, it tells you which device is involved, where it is and who depends on it.
- The **CIS** (Center for Internet Security) Controls ask for up-to-date architecture diagrams:
 - [Safeguard 12.4 in the Controls Navigator](#)
- The **NIST** (National Institute of Standards and Technology) Cybersecurity Framework asks for maps of network communications and data flows (subcategory **ID.AM-03** (Identify function, Asset Management category, item 03)).
- For continuous monitoring programs, see the **NIST** (National Institute of Standards and Technology) guide:
 - [Special Publication 800-137](#)



Documentation is sensitive

- A complete diagram tells an attacker exactly where to go.
- Store it with restricted access; never post it.
- Never use a real organization's documents in course work without written permission (Legal Notice, sections 7.5 and 7.9).

Types of network documentation (1/2)

Document	What it shows	How the monitoring technician uses it
Asset inventory	Every device: hostname, address, family, owner, location, dependent devices, priority	Decide what to monitor and in what order; recognize unknown devices
Physical network diagram	Rooms, racks, devices, cables, switch ports, wall jacks	Locate a device, trace a cable, find where to connect a sensor
Logical network diagram	Networks, address ranges, VLAN (Virtual Local Area Network) numbers, gateways, firewall zones	See which devices may communicate; find the trust boundaries
IP (Internet Protocol) address plan	Which address belongs to which device and network; often kept in an IPAM (Internet Protocol Address Management) tool	Identify the device behind an address found in a log or an alert
Cable map (port map)	Which device is connected to which switch port	Find the port to mirror, or the port to disable during an incident



Types of network documentation (2/2)

Document	What it shows	How the monitoring technician uses it
Rack diagram	Position of each device in each rack	Guide a person on site to the right device
Data-flow diagram	Which systems exchange data, in which direction, for what purpose	Know which traffic is normal (the baseline) and which is unexpected
Configuration records	Saved settings of each device	Detect unauthorized changes by comparison
Change log	What changed, when, by whom	Explain a sudden change in the monitoring data
Contacts and service agreements	Who owns what; what the provider guarantees (SLA (Service Level Agreement))	Know whom to call and which response time to expect



For every document, ask two questions: is it current (version and date)? Who owns it?

Physical and logical diagrams



Physical diagram

Where is it, and how is it connected?

- Rooms, racks and the position of each device in its rack
- The switch port of each cable and the wall jack at the other end
- The type of link: copper, fibre or radio
- **Use it to:** find a device, trace a cable, know which port to disconnect in an emergency



Logical diagram

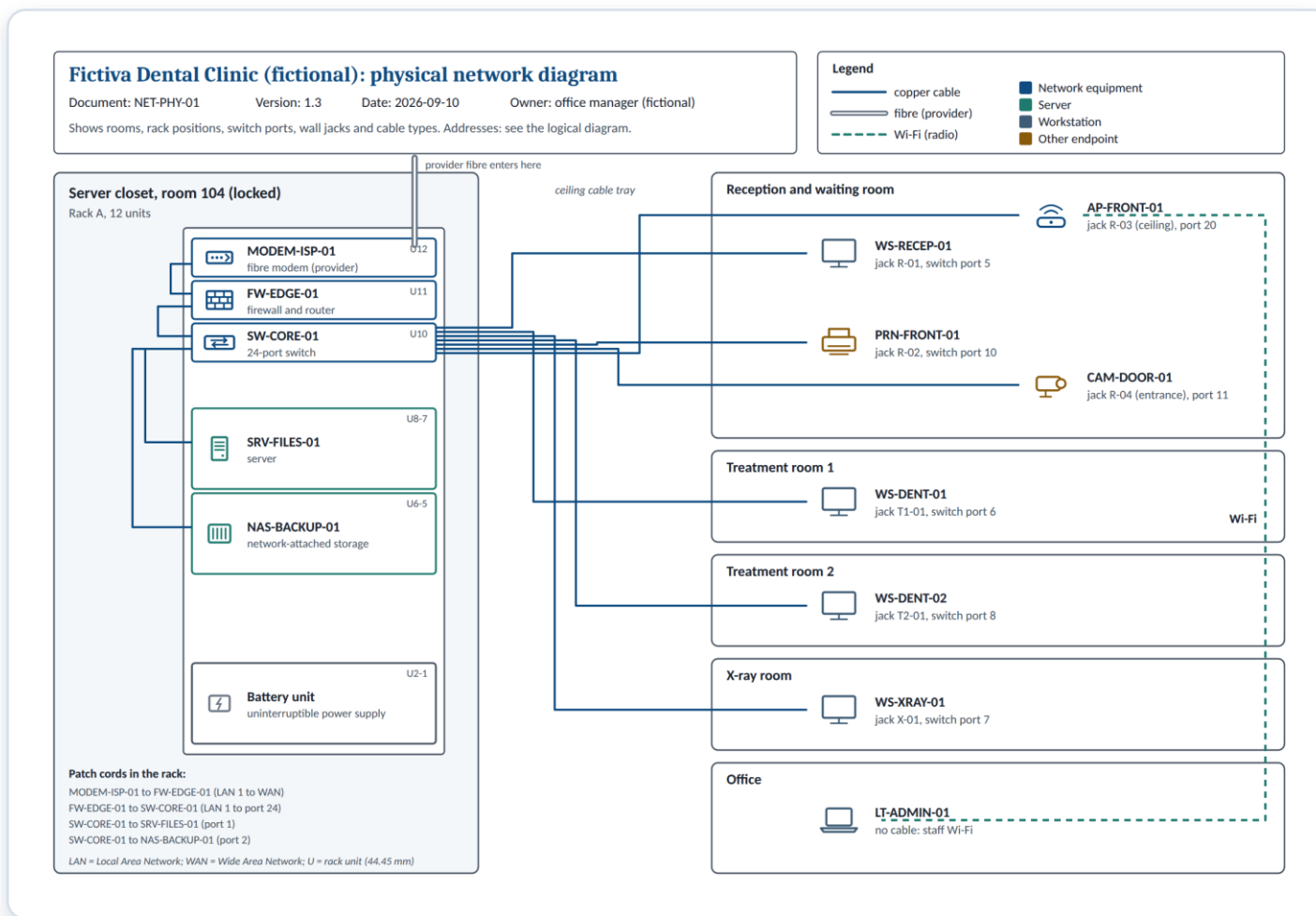
Which network is it in, and what may it communicate with?

- Networks, address ranges and **VLAN** (Virtual Local Area Network) numbers
- Default gateways, firewalls between zones, allowed flows
- It ignores rooms and racks
- **Use it to:** see which devices may communicate, find the **trust boundary**, choose where to observe traffic



Both start with a **title block** (name, version, date, owner) and a **legend** that explains symbols and line styles. A diagram without a date cannot be trusted.

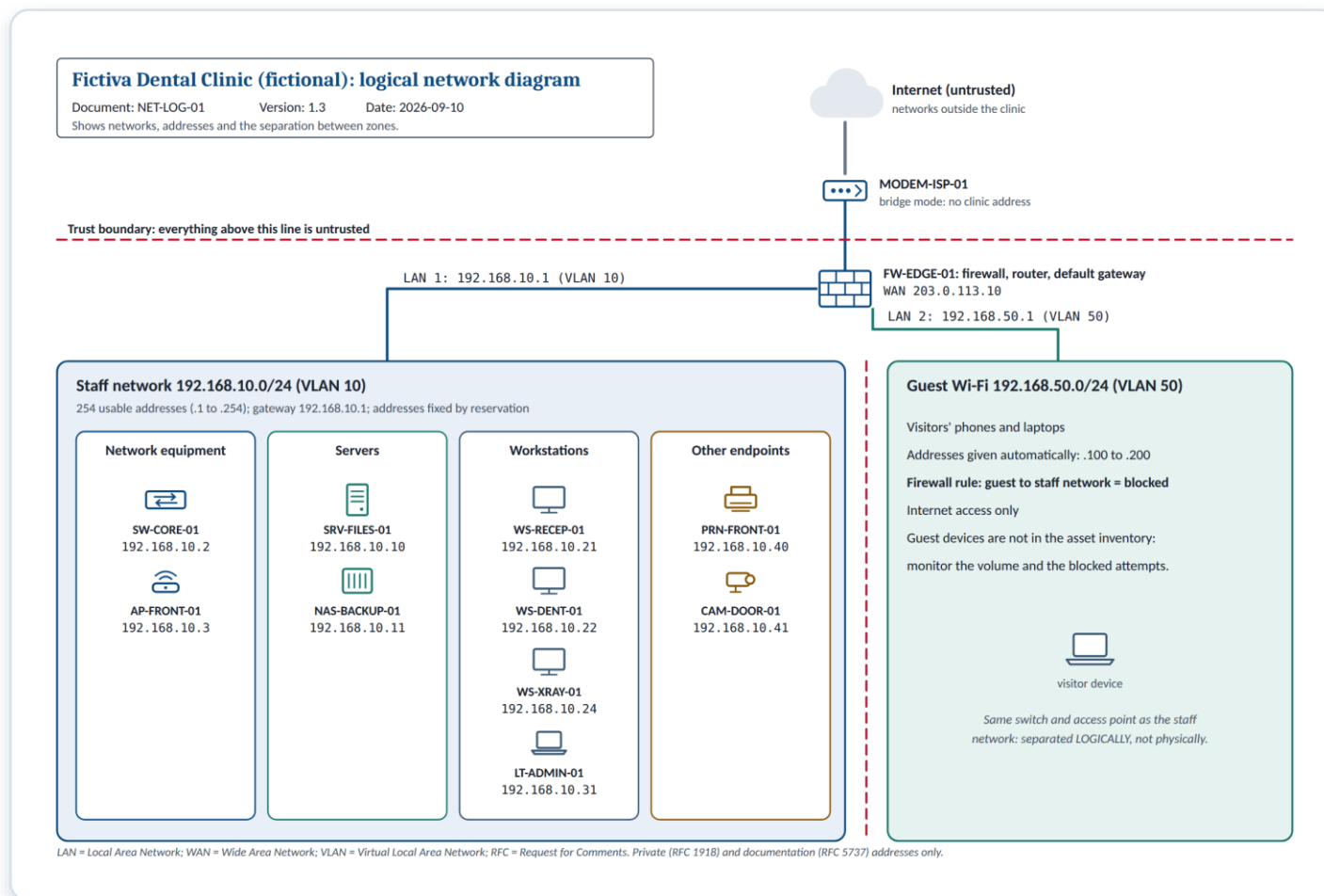
Physical diagram of the clinic



How to read it

- WS-DENT-01 is in treatment room 1, on wall jack T1-01.
- The jack is cabled to port 6 of SW-CORE-01, in the server closet.
- The switch reaches the firewall through port 24.
- LT-ADMIN-01 has no cable: it reaches the network through AP-FRONT-01.
- Find WS-DENT-02 on port 8. Is it in the inventory?

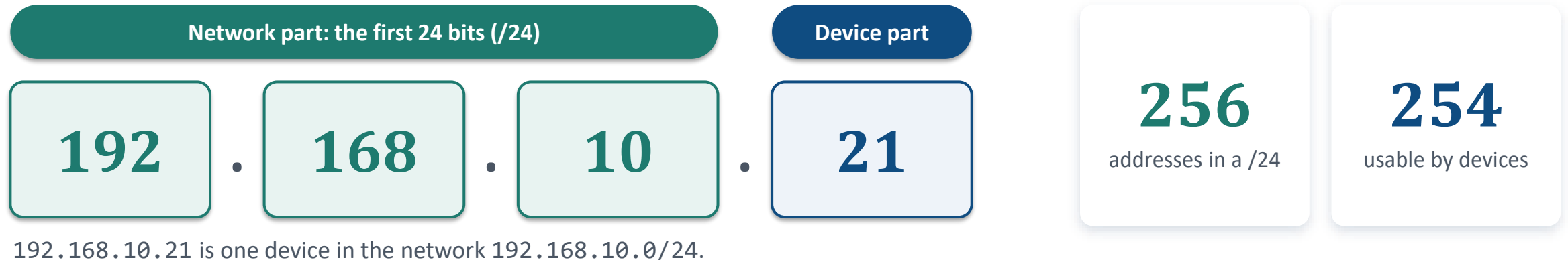
Logical diagram of the clinic



How to read it

- Staff network: 192.168.10.0/24, default gateway 192.168.10.1 on FW-EDGE-01.
- Guest Wi-Fi: 192.168.50.0/24, a separate network.
- Both use the same switch and access point: only the logical diagram shows the separation.
- The firewall blocks guest traffic toward the staff network.

Reading the addresses (1/2)



- An **IPv4** (Internet Protocol version 4) address is a 32-bit number written as four numbers from 0 to 255.
- A **subnet** is a range of consecutive addresses that form one network, written in **CIDR** (Classless Inter-Domain Routing) notation: 192.168.10.0/24.
- In a /24, .0 identifies the network, .255 is the broadcast address, and .1 to .254 go to devices.
- The same prefix written as a **subnet mask**: 255.255.255.0.

Reading the addresses (2/2)

- The **default gateway** is the router interface inside the subnet: 192.168.10.1 on FW-EDGE-01. Every packet for another network goes to it.
- **Private addresses**, defined in [Request for Comments 1918](#): 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. Used inside organizations, never routed on the Internet.
- **Documentation addresses**, reserved by [Request for Comments 5737](#): 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24. For examples only.
- A **VLAN** (Virtual Local Area Network) is a logical network created on switches and access points. Devices in different **VLAN** (Virtual Local Area Network) segments exchange data only through a router or a firewall.



30-second exercise on Figure 2.2-B

1. Which address is the default gateway of CAM-DOOR-01?
2. Could a visitor's phone on the guest Wi-Fi reach SRV-FILES-01?

Answer

1. 192.168.10.1, the staff interface of FW-EDGE-01.
2. No: the phone is in 192.168.50.0/24, a different subnet, and the firewall blocks guest traffic toward the staff network.



Topologies

- The **topology** is the arrangement of nodes and links: the **physical topology** (cabling) and the **logical topology** (how data travels).
- Common shapes: star, extended star, mesh, point-to-point and hybrid. Bus and ring are older and rarely installed.
- Fictiva is an **extended star**: a star around SW-CORE-01, and a small star around AP-FRONT-01 for Wi-Fi devices.
- It is also a **tree topology**: connected, with no loop, so there is exactly one path between any two devices.
- Consequence: every device in the middle of a path is a single point of failure for that path.

14

devices in the documentation

13

links (cables and radio)

1

path between any two devices



For monitoring, the shape shows where traffic converges (where to observe it) and where one failure stops many devices.



Common topologies

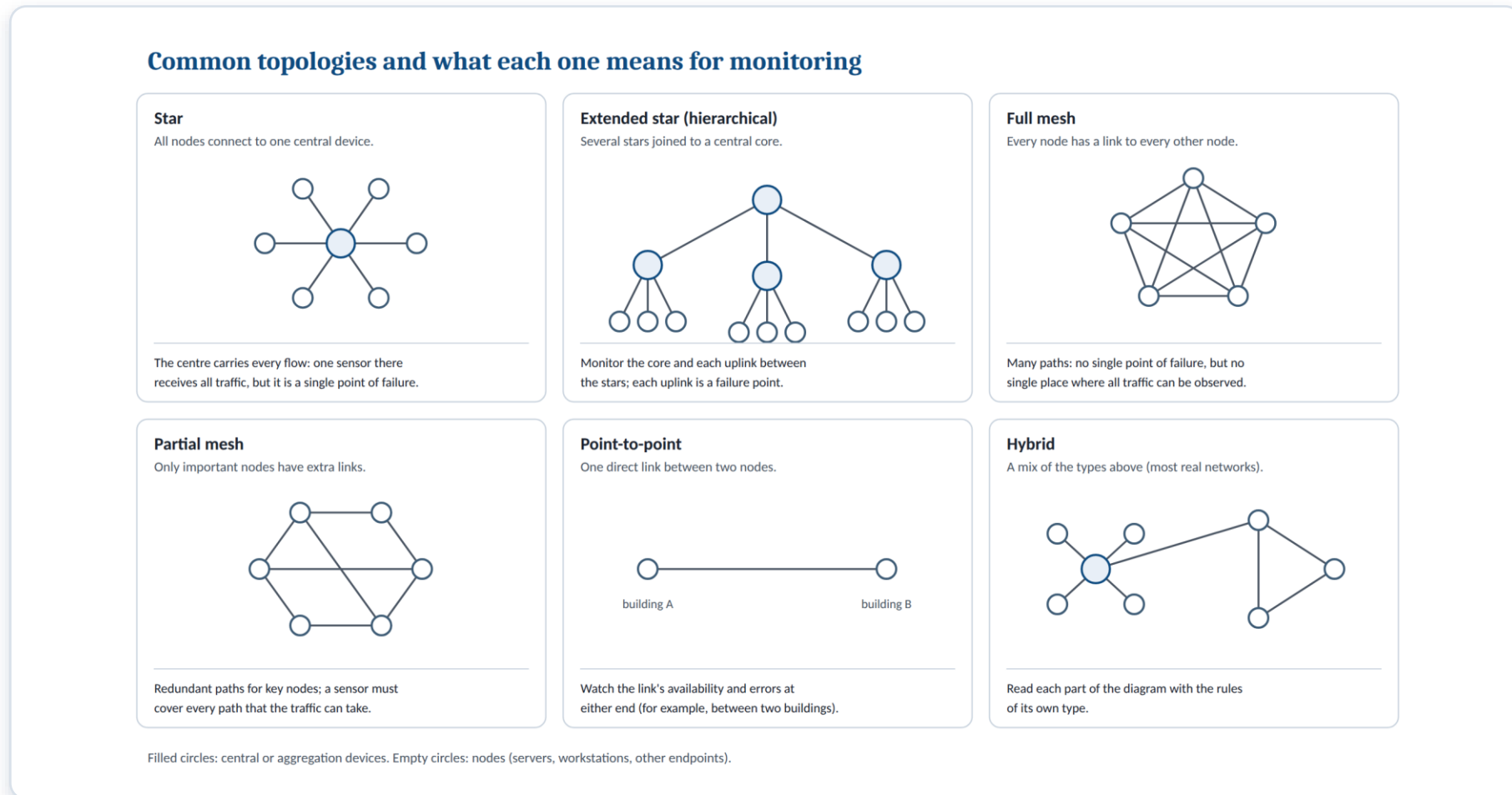


Figure 2.2-C. Common topologies, with what each one means for monitoring.

A method to read any network diagram

1

Is the document current?

Title block: name, version, date, owner

2

What do the symbols and lines mean?

The legend

3

Where does the network end, and what are its zones?

Internet, provider, staff, guests, servers

4

Which devices exist, in which family?

Symbols and hostnames, compared with the inventory

5

Which networks and addresses are used?

Subnets, gateways, **VLAN** (Virtual Local Area Network) numbers

6

How does each important flow travel?

Follow the lines from source to destination

7

Where do flows converge? What has no backup?

Devices crossed by many flows; single links

8

What is missing or contradictory?

Differences with the other documents



Steps 6 and 7 are exactly what the script of example 2.2 automates.



When documentation and reality disagree

Documentation starts to age on the day it is written: devices are added in a hurry, moved, replaced or removed.



Why it matters

- A device missing from the documents is not monitored.
- A removed device that is still documented produces false “device down” alerts.



What a technician does

- Compare the documents with each other.
- With the owner's authorization, compare them with what the network reports (switch address table, **DHCP** (Dynamic Host Configuration Protocol) leases).
- Record each difference; ask the owner; update the documents with a new version and date.



What a technician never does

- Fill the gap by scanning a network without written authorization: a scan is recorded as an attack (Legal Notice, section 7.6).
- Lab 01 involves no scanning at all.



Fictiva: the cable map lists WS-DENT-02 on switch port 8, but the inventory has no WS-DENT-02. A forgotten computer, or a removed computer and an outdated cable map? The script of example 2.2 finds this difference by itself.

Videos for Lesson 2.2 (after class)



Network Documentation

Professor Messer · **about 10 min**

CompTIA (Computing Technology Industry Association)
Network+ N10-009, objective 3.1

Watch for: physical and logical network maps, rack diagrams, cable maps, layered diagrams, asset tags and the asset database, address management, site surveys.

Watch on YouTube

Page with transcript



Network Topologies

Professor Messer · **about 6 min**

CompTIA (Computing Technology Industry Association)
Network+ N10-009, objective 1.6

Watch for: star (hub-and-spoke), mesh, hybrid, spine-and-leaf and point-to-point designs, and why redundancy changes what a single failure does.

Watch on YouTube

Page with transcript



Links checked on September 20, 2026. Third-party content, no affiliation with Cyber.SoHo. Check each link with [VirusTotal](#) first.



Example 2.2: where to observe the traffic



Task

Use the documentation packet (cable map, inventory, data flows) to follow each flow, find the device that carries the most flows (the best place for a sensor) and the single points of failure, and check the cable map against the inventory.



Link to the lecture

Applies 2.2.2 (physical connections), 2.2.4 (tree topology, single points of failure), steps 6 and 7 of the reading method, and 2.2.6 (differences between documents).



Link to the labs

Lab 01: your plan says where traffic will be observed. **Lab 02:** the sensor goes on a mirror port of the core switch; the normal volume of each flow becomes the baseline. **Lab 03:** the paths show which devices an anomaly crossed.



The situation

The office manager adds two documents to the packet: the installer's cable map, and the list of data flows the clinic uses every day.

“Where should we watch our traffic, and what happens if one box fails?”

The documentation packet

Cable map: 13 links

Device A	Device B	Medium
INTERNET	MODEM-ISP-01	fibre
MODEM-ISP-01	FW-EDGE-01	copper
FW-EDGE-01	SW-CORE-01	copper
SW-CORE-01	SRV-FILES-01	copper
SW-CORE-01	NAS-BACKUP-01	copper
SW-CORE-01	WS-RECEP-01	copper
SW-CORE-01	WS-DENT-01	copper
SW-CORE-01	WS-XRAY-01	copper
SW-CORE-01	WS-DENT-02	copper
SW-CORE-01	PRN-FRONT-01	copper
SW-CORE-01	CAM-DOOR-01	copper
SW-CORE-01	AP-FRONT-01	copper
AP-FRONT-01	LT-ADMIN-01	wireless

Documented data flows: 10

Flow	Source	Destination	Purpose
1	WS-RECEP-01	SRV-FILES-01	Patient records
2	WS-DENT-01	SRV-FILES-01	Patient charts
3	WS-XRAY-01	SRV-FILES-01	X-ray images
4	LT-ADMIN-01	SRV-FILES-01	Shared folders
5	SRV-FILES-01	NAS-BACKUP-01	Nightly backup
6	CAM-DOOR-01	NAS-BACKUP-01	Video recording
7	WS-RECEP-01	PRN-FRONT-01	Printing
8	WS-RECEP-01	INTERNET	Email and insurers' portal
9	LT-ADMIN-01	INTERNET	Email and web
10	SRV-FILES-01	INTERNET	Software updates



Ports and wall jacks are on Figure 2.2-A. Spot the difference: WS-DENT-02 is in the cable map, not in the inventory.

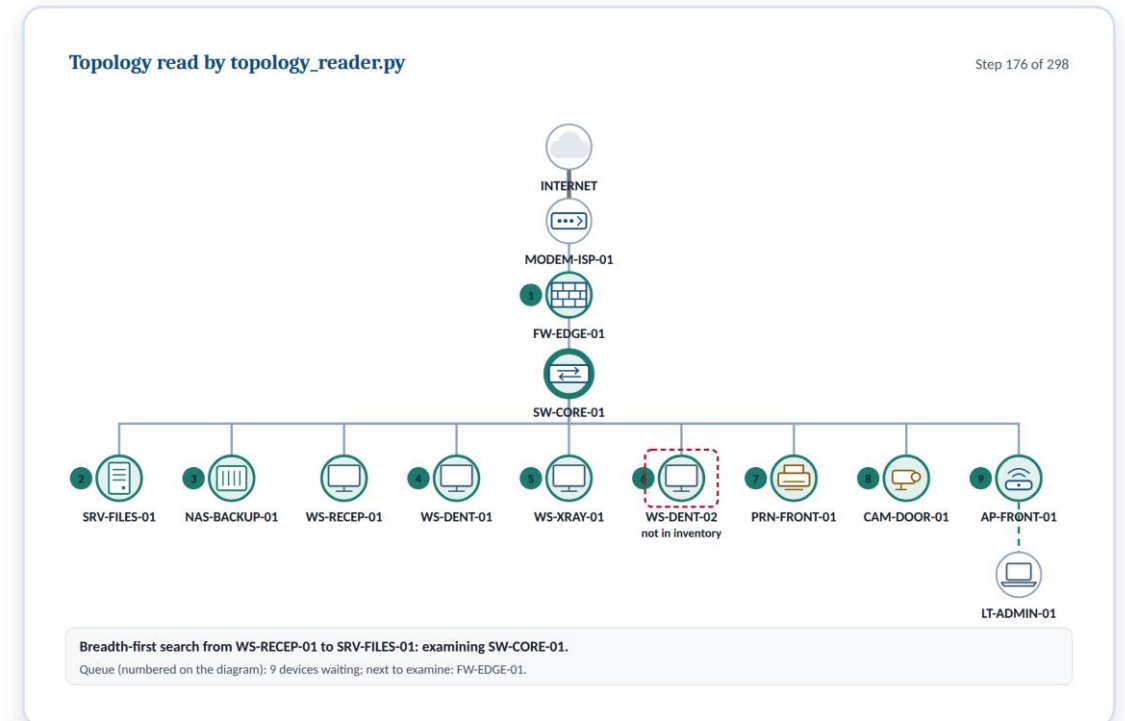
Code 2.2: topology_reader.py

What the script does

1. Reads the cable map into a table of neighbours: each cable works in both directions.
2. Compares it with the inventory: WS-DENT-02 is not in the inventory.
3. Checks the shape: 13 links for 14 devices, so no loop.
4. `find_path()` runs a breadth-first search for each of the 10 flows.
5. Counts the flows through each device, ranks them and saves `monitoring_points.csv`.

```
$ python3 topology_reader.py  
# Windows: py topology_reader.py
```

Run it only in your lab environment, next to both CSV files.



Live visual in the lecture page: the search from WS-RECEP-01 examines SW-CORE-01; nine devices wait in the queue (numbered).

Breadth-first search in code

topology_reader.py

```
def find_path(graph, start, goal):
    came_from = {start: None}           # device -> device we came from
    queue = deque([start])              # waiting line of devices to examine
    while queue:                        # repeat while a device is waiting
        current = queue.popleft()        # take the first device out
        if current == goal:              # destination reached: stop
            break
        for neighbour in graph[current]: # every connected device
            if neighbour not in came_from: # not reached before
                came_from[neighbour] = current
                queue.append(neighbour)    # examine it later
    if goal not in came_from:            # no documented connection
        return []
    path = [goal]                       # rebuild the path backwards
    while path[0] != start:
        path.insert(0, came_from[path[0]])
    return path
```

- A **breadth-first search** examines every device one link away, then two links away, and so on.
- queue: devices waiting to be examined, first in, first out.
- came_from: for each device reached, the device just before it.
- The path is rebuilt backwards, from the goal to the start.
- It works on any topology, including networks with several paths.

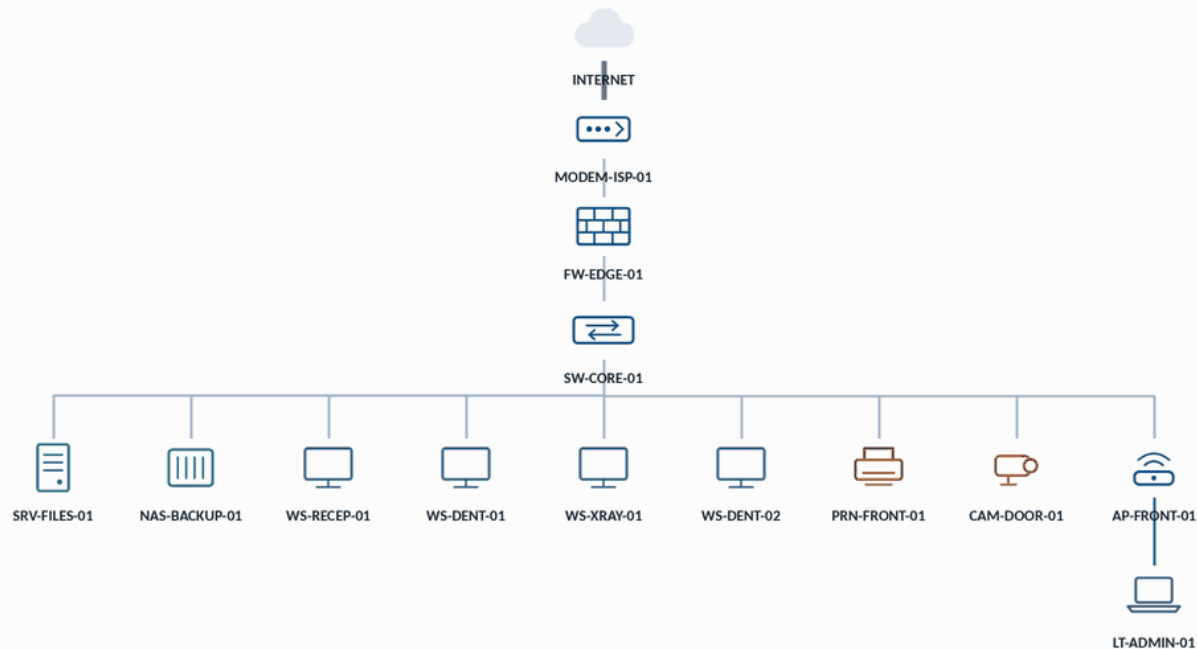
Condensed excerpt. The full script, with a comment before every line, is in the lecture page and the download package.



Following the ten flows

Following the documented flows through the Fictiva topology

Start



10 documented data flows will be followed, one at a time.

Each device in the middle of a path gets +1: the badge shows how many flows it carries.

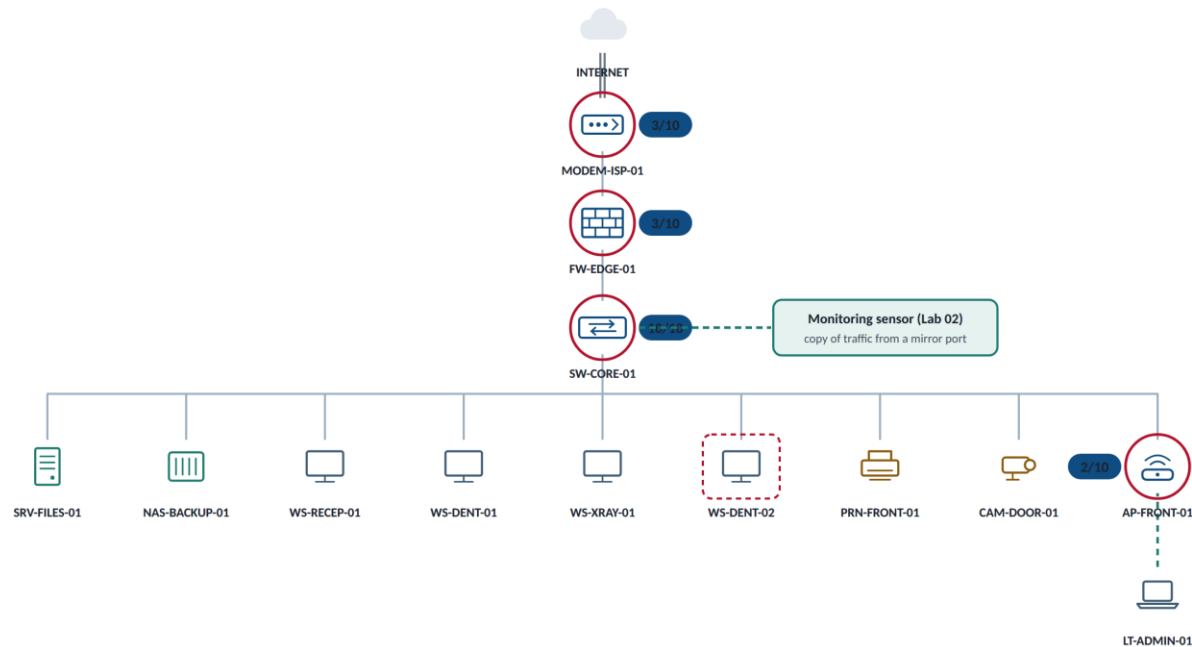
- Each documented flow is highlighted on its path, one after the other.
- Every device in the middle of a path receives +1.
- The counters grow on SW-CORE-01, FW-EDGE-01, MODEM-ISP-01 and AP-FRONT-01.
- The last frame shows the best monitoring point and the single points of failure.

Result: where to observe, what to protect

What the topology tells the monitoring plan

Blue badge = number of documented flows (out of 10) that pass through the device.

Red ring = single point of failure (only one path exists). Dashed red box = in the cable map, not in the inventory.



SW-CORE-01 carries 10 of 10 flows: a mirror port on it gives the sensor a copy of all documented flows. FW-EDGE-01 alone would show only the 3 Internet flows.

SW-CORE-01

10/10 flows

FW-EDGE-01

3/10 flows

MODEM-ISP-01

3/10 flows

AP-FRONT-01

2/10 flows

- A mirror port on SW-CORE-01 gives the sensor a copy of every documented flow (Lab 02).
- Monitoring only at the firewall would miss the 7 internal flows.
- All four are single points of failure. WS-DENT-02: confirm with the clinic.



Fictiva_Topology_Documentation.xlsx: same analysis, with formulas you can check.



Check yourself: flash cards 2.2

1

How do a physical and a logical diagram differ?

Physical: where devices are and how they are cabled (rooms, racks, ports, jacks, cable types). Logical: networks, address ranges, **VLAN** (Virtual Local Area Network) numbers, gateways and zones, whatever the location.

2

What is a topology?

The arrangement of the nodes and links of a network. Physical topology: the cabling. Logical topology: how data travels. Common shapes: star, extended star, mesh, point-to-point, hybrid.

3

What is a single point of failure?

A component whose failure stops a function for the devices that depend on it, because no other path or backup exists. Example: the only core switch of a star topology.

4

What does 192.168.10.0/24 mean?

A network whose first 24 bits are fixed (mask 255.255.255.0): addresses 192.168.10.0 to 192.168.10.255, with 254 usable device addresses, .1 to .254.

5

Why must documentation be compared with reality?

It ages: a device missing from the documents is not monitored, and a removed device produces false alerts. Each difference is a finding to confirm with the owner.

Practice questions 2.2 (1/3)

Question 1 · one answer

Which document tells you that WS-DENT-01 is on wall jack T1-01 and on port 6 of SW-CORE-01?

- a) The logical network diagram
- b) The physical network diagram, or its cable map
- c) The address plan
- d) The data-flow diagram

Answer: b

Rooms, racks, cables, ports and wall jacks are physical information. The logical diagram shows networks, addresses and zones.

Question 2 · one answer

Guest Wi-Fi 192.168.50.0/24 and the staff network 192.168.10.0/24 appear as two networks, although both use AP-FRONT-01 and SW-CORE-01. Why?

- a) The diagram is wrong: one access point serves one network
- b) A logical diagram shows networks and their separation, not shared boxes
- c) Guest devices use a different cable
- d) The access point is really two devices

Answer: b

The separation is logical: **VLAN** (Virtual Local Area Network) settings on the switch and access point, and rules on the firewall.



Practice questions 2.2 (2/3)

Question 3 · one answer

What does the notation 192.168.10.0/24 mean?

- a) The network contains 24 devices
- b) The first 24 bits identify the network; 256 addresses, 254 usable
- c) The network starts at address 192.168.10.24
- d) Device number 24 is the default gateway

Answer: b

/24 equals the mask 255.255.255.0; .0 identifies the network and .255 is the broadcast address.

Question 4 · one answer

The cable map lists WS-DENT-02 on switch port 8, but the inventory has no WS-DENT-02. What do you do first?

- a) Ignore it: cable maps are often wrong
- b) Scan the whole network to find the computer
- c) Record the difference and ask the owner which document is correct
- d) Disable port 8 on the switch

Answer: c

A documentation difference is a finding: the owner confirms, then the documents are updated. Scanning needs written authorization (section 7.6).



Practice questions 2.2 (3/3)

Question 5 · 4 answers

Which elements belong on a well-made network diagram? Select all that apply.

- a) A legend for symbols and line styles
- b) The hostnames of the devices
- c) Network addresses and gateways
- d) A title block: version, date, owner
- e) The administrator passwords
- f) Every device's full configuration

Answer: a, b, c, d

Keep what helps a reader understand the network. Passwords belong in a protected password manager; configurations in the configuration records.

Question 6 · 4 answers

A sensor on a mirror port of SW-CORE-01 receives a copy of which traffic? Select all that apply.

- a) Reception desktop to the file server
- b) File server to the backup storage
- c) Office laptop to the Internet
- d) A call on a staff member's cellular plan
- e) Entrance camera to the backup storage
- f) Traffic inside the provider's network

Answer: a, b, c, e

Follow each path: every internal and Internet flow of the clinic crosses SW-CORE-01. Cellular traffic and the provider's network never do.



Lab 01 preview (summary only)



Lab 01 (5%): map a small network and write its monitoring plan. Friday, September 25, 2026 · Class 7 · online. The lab guide, the documentation and the grading grid are given in class.

What you will do

- Receive the documentation of a small fictional network.
- Build its asset inventory: hostnames, addresses, families, dependent devices, priorities (2.1).
- Read and complete its physical and logical diagrams (2.2).
- Identify its data flows, candidate monitoring points and single points of failure (2.2).
- Write a first monitoring plan: what, where, in what order (2.3 to 2.5 come in Classes 6 and 7).

How the three labs connect

1

Part 1 · Lab 01 · September 25

The inventory, the diagrams and the monitoring plan

2

Part 2 · Lab 02 · October 3

Install the tool at the chosen point; build the baseline; set the alerts

3

Part 3 · Lab 03 · October 16

Detect, diagnose and report an anomaly; propose improvements



Before Lab 01: sign Annex A of the [Cyber.SoHo Legal Notice](#) for this course. Documentation work only: no scanning, no traffic capture, no real network.

Key points and next class

- ✓ You cannot monitor what you do not know: the asset inventory comes first.
- ✓ The family depends on the role: servers serve, people use workstations, network equipment forwards other devices' traffic, other endpoints run firmware.
- ✓ Switches decide with **MAC** (Media Access Control) addresses (layer 2); routers and firewalls with **IP** (Internet Protocol) addresses (layer 3).
- ✓ Physical diagram: where, and how connected? Logical diagram: which network, and what may communicate?
- ✓ Following the flows shows where to observe traffic and which devices are single points of failure.
- ✓ Every documentation difference is a finding to confirm with the owner, never a reason to scan without authorization.



Next class

Thursday, September 24, 2026 (Class 6, online): 2.3 Organizational and operational constraints of the network; 2.4 Identifying monitoring needs.

To prepare: watch the four videos; run both scripts in your lab environment; open both workbooks and change one value.

Quiz 2: Saturday, September 26 (5%), covers all material to date.

Links used in this lecture (1/2)



[Cyber.SoHo Legal Notice, version 2.0](#)

Terms of use and acceptable use policy for all Cyber.SoHo materials



[Critical Security Controls Navigator](#)

CIS (Center for Internet Security): the 18 Controls and their safeguards



[Inventory and Control of Enterprise Assets](#)

Control 1 of the CIS (Center for Internet Security) Controls



[Cybersecurity Framework](#)

Published by the NIST (National Institute of Standards and Technology)



[Special Publication 800-137: continuous monitoring](#)

Published by the NIST (National Institute of Standards and Technology)



[Baseline cyber security controls](#)

Canadian Centre for Cyber Security, small and medium organizations



[Ethernet and Wi-Fi standards committee \(802\)](#)

The IEEE (Institute of Electrical and Electronics Engineers)



[Request for Comments 1918](#)

Private address ranges used inside organizations



[Request for Comments 5737](#)

Address ranges reserved for documentation



[Request for Comments 2606](#)

Reserved domain names such as .example



[Request for Comments 7042](#)

Includes hardware addresses reserved for documentation



[Criminal Code, section 184](#)

Interception of private communications



[Criminal Code, section 342.1](#)

Unauthorized use of a computer



[Law 25: private-sector privacy act \(P-39.1\)](#)

Protects personal information in Québec



[Commission d'accès à l'information du Québec](#)

Supervises the protection of personal information



Links used in this lecture (2/2)



[What is a Server? Servers vs Desktops Explained](#)

Video, PowerCert Animated Videos, 7 min 14 s (Lesson 2.1)



[Hub, switch or router? Network devices explained](#)

Video, PieterExplainsTech, about 8 min (Lesson 2.1)



[Network Documentation \(N10-009, 3.1\)](#)

Video and transcript, Professor Messer, about 10 min (Lesson 2.2)



[Network Topologies \(N10-009, 1.6\)](#)

Video and transcript, Professor Messer, about 6 min (Lesson 2.2)



[Free Network+ N10-009 course](#)

Professor Messer: complete index of free networking videos



[Python downloads](#)

Official source of the Python interpreter for the two scripts



[draw.io \(diagrams.net\)](#)

Free diagram editor for the diagrams of Lab 01



[Oracle VirtualBox](#)

Free virtualization software for your lab environment



[Wireshark](#)

Packet analyzer used in Topic 4, on lab traffic only



[VirusTotal: check a link](#)

Check every link before opening it (Legal Notice, Annex C)



[Google Safe Browsing site status](#)

Second check of a domain's reputation



[Justice Laws Website](#)

Official consolidated text of Canadian federal laws



[LégisQuébec](#)

Official consolidated text of Québec laws



[Institutional Student Evaluation Policy](#)

Dawson College policy that governs all assessments



[Bibliothèque et Archives nationales du Québec](#)

Free library account: O'Reilly and LinkedIn Learning resources



Thank you

Questions? Ask now in the class chat, or through the College's messaging system after class.

Next: Thursday, September 24, 2026, Class 6 (online).

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- Fictional scenario: private (**RFC** (Request for Comments) 1918) and documentation (**RFC** (Request for Comments) 5737) addresses, reserved domain (**RFC** (Request for Comments) 2606).
- Scripts only in your Authorized Lab Environment (sections 6.1 to 6.4 and 7.2). See [Criminal Code, section 184](#), [section 342.1](#) and [Law 25](#).
- Full text: [Cyber.SoHo Legal Notice, version 2.0](#).

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